



Estd: 1986

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SJC INSTITUTE OF TECHNOLOGY

Chickballapur – 562 101

Department of Artificial Intelligence and Data Science

Tutorial - 3

SUBJECT TITLE	Big Data Analytics		
SUBJECT TYPE	Integrated		
SUBJECT CODE	BAD601		
ACADEMIC YEAR	2024-25 (Even semester)	Batch	
SCHEME	NEP CBCS 2022		
SEMESTER	6th sem, AI&DS		
FACULTY NAME and DESIGNATION	Darshan A, Assistant Professor		
DATE of ISSUED	23-05-2025		

Q. No.	Questions	Level	COs
1	Define a) Datawarehousing b) Hive List out and explain the features and uses of Hive.	L2	CO4
2	List out and explain the hive data units. And also explain the hive storage structure.	L2	CO4
3	List out and explain types of metastore. Explain the concept of partitioning and types of it with example queries.	L2	CO4
4	Explain hive architecture with a neat block diagram.	L2	CO4
5	List out different data types of hive. Explain different hive file format. Explain the concept of joins in hive with example.	L2	CO4
6	Write a hive query for the below given statements: a) Write a query to create a database named "STUDENTS" with comments and database properties. b) Write a query to list out all databases and describe it. c) Write a query to alter the "Students" database, i.e change the database properties i.e change the database modified date to 23-05-2025. d) write a query to drop a database named "Students" which has the tables within it. e) write a query to display the databases that are having a name beginning with 'm' and has three letters and ends with 'g'.	L3	CO4
7	Explain different types of tables with example. Write a query for loading data into table from a file. a) Write a query to retrieve the student details from "EXT_STUDENT"	L3	CO4

	table. b) write a query to create a view table which should have rollno and name from the table EXT_STUDENT. And then limit the records to 4 and then drop the view table.		
8	Define pig. List out the features of pig. And also explain the pig philosophy. And also explain ETL processing in pig. When to use pig and when not to use pig?	L2	CO4
9	Write the queries for the given problem statements: a) Find the tuples of those students where the GPA is greater than 4.0 b) Display the names of all the students in upper case. c) Group tuples of students based on their GPA. d) Remove duplicate tuples of students e) display the top three students name and roll number based on their marks f) Arrange all the students in descending order based on rollno. h) Join two relations namely, 'student' and 'department' based on the values contained in the "rollno" column. i) Divide the relation based on the GPA's acquired by the students: - GPA > 4.0, Place it into relation X. - GPA < 5.0, place it into relation Y.	L3	CO4
10	Explain below given with example queries: a) sample b) avg c) max d) min e) sum f) count	L2	CO4
11	Explain complex data types in pig with sample queries demonstrating each.	L2	CO4
12	Write a pig query to demonstrate the word count. Differentiate between PIG vs HIVE	L3	CO4
13	Define spark. Explain the architecture of the spark with the neat diagram.	L2	CO5
14	List out and explain the main features of spark.	L2	CO5
15	Explain the spark software spark components with the neat diagram.	L2	CO5
16	List out and explain the steps between acquisition of data from different sources and its applications.	L2	CO5
17	Explain the working of SparkSQL and its features.	L2	CO5
18	Write a program to implement a word count program in spark.	L3	CO5
19	Write a program using pig latin scripts to sort, group, join, project, and filter the data.	L3	CO4
20	Write a program using Hive to create, alter, and drop databases, tables, views, functions and indexes.	L3	CO4

Sign of the faculty
Darshan A

HOD